

create dsa notes

Data Structures 1. Introduction Data Structures are methods to organize data in a way that allows for efficient storage and retrieval. They are used to manage large amounts of data and provide a way to perform operations on the data. 2. Types of Data Structures Basic: Arrays: elements stored in continuous memory Linked List: each node points to the next node Stack: LIFO (Last In First Out) structure Queue: FIFO (First In First Out) structure Advanced: Trees: hierarchical structure Graphs: nodes connected by edges Hash Table: key-value storage 3. Operations Insertion: add an element to the data structure Deletion: remove an element from the data structure Search: find an element in the data structure Traversal: visit all elements in the data structure 4. Importance Efficient memory usage Fast access to elements Better performance in terms of time and space complexity
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